

Felipe Ramos Roque

Applied AI Engineer | AI Agents & Multi-Agent Systems | Process Automation | Python, TypeScript, LangGraph

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Open to remote roles worldwide - full overlap with US business hours - available as an international contractor

PROFESSIONAL SUMMARY

Applied AI engineer with 5+ years building production automation for complex business processes. Co-built and operates a multi-tenant AI agent platform serving 22 client brands in production on Kubernetes, architected on LangGraph with hybrid retrieval, guardrails, and end-to-end tracing. Having delivered 20+ production RPA automations before moving into agent architecture, brings both sides of the enterprise automation shift - the process knowledge to understand what actually breaks in a business workflow, and the engineering to replace it with reliable, observable multi-agent systems. Automation delivered across multiple business fronts has removed over 500 hours of manual processing per month, handling tens of thousands of records monthly. Postgraduate in Artificial Intelligence for Developers. Fluent in English and Portuguese.

TECHNICAL SKILLS

AI Agents & Orchestration: AI Agent Development, Multi-Agent Systems, Agent Orchestration, LangGraph (StateGraph), LangChain, Tool Use & Function Calling, Memory & State Management, Conversation Checkpointing, LLM Integration (OpenAI, Anthropic, Azure OpenAI), Prompt Engineering, AI Guardrails

Retrieval & RAG: Retrieval-Augmented Generation (RAG), Hybrid Retrieval (BM25 + FAISS), Reciprocal Rank Fusion, Cross-Encoder Reranking, Vector Stores (FAISS, Chroma), Embeddings, Multi-Tenant Knowledge Bases

Evaluation & Observability: Agent Tracing (Langfuse, LangSmith), OpenTelemetry, Behavioral Testing, Prompt & Configuration Versioning, Monitoring, Error Handling, Operational Dashboards

Software Engineering: Python, TypeScript, JavaScript, Java, SQL, FastAPI, Node.js, React, Next.js, REST API Design & Integration, Automated Testing, CI/CD Pipelines, Git

Process Automation: RPA (Robotic Process Automation), Business-Process Reengineering, Process Discovery & Documentation, n8n, Workflow Orchestration, Selenium, OpenCV, Web Scraping

Enterprise Integration: CRM Platforms, ERP Systems, Warehouse Management (WMS), Ticketing Systems, Data Warehouse Integration, WhatsApp Business Platform, Meta Graph API, Webhooks

Cloud & Data: Kubernetes, Docker, Multi-Tenant Architecture, Cloud Deployment, GitLab CI, Azure OpenAI Service, PostgreSQL, MongoDB, Redis, Pandas, ETL

Languages Spoken: Portuguese (Native), English (Fluent)

PROFESSIONAL EXPERIENCE

AI Engineer

BasicX Solutions - São Paulo, SP, Brazil | October 2025 - Present

- Co-build and operate a multi-tenant conversational AI platform on WhatsApp, deployed as isolated Kubernetes services for 22 client brands across healthcare, veterinary, retail marketplace, real estate, fitness, and beauty verticals.
- Engineered the core agent as a hand-built LangGraph StateGraph - intent classification, lead extraction and persistence, retrieval, profile loading, a tool-calling agent loop, an LLM supervisor, and human-handoff detection - with checkpointed conversation state.
- Built a hybrid retrieval layer combining BM25 and FAISS through Reciprocal Rank Fusion with cross-encoder reranking, over per-tenant isolated vector stores.
- Designed a routing-agent and named-specialist architecture, with up to nine domain specialists per tenant, each bound to its own knowledge base and tool set.
- Implemented a supervisor and guardrail layer enforcing brand voice, output format, age gating, and crisis-trigger detection, backed by a behavioral test suite.
- Instrumented production agents with Langfuse and LangSmith tracing over OpenTelemetry, and hardened them with concurrency tests covering multi-replica handoff locking and message debouncing.
- Built configuration as a product: Pydantic-validated per-tenant YAML with a web builder providing versioning and diffing, plus a CI job asserting deployed configuration matches the repository.
- Delivered supporting automation - an ERP-to-WMS order fulfillment integrator, a deterministic payroll-billing audit engine with a human approval gate, and migrations of n8n no-code workflows into tested Python services.

Founder & Principal Consultant

FEROQ - Worldwide | 2021 - Present

- Run an independent technology consultancy delivering AI agents, intelligent process automation, and full-stack applications to clients across healthcare, finance, and logistics.
- Own the full delivery lifecycle with minimal supervision: requirements discovery, solution architecture, development, cloud deployment, monitoring, and ongoing client support.
- Build and maintain applications in Python, Node.js, TypeScript, and React with automated test suites and CI/CD pipelines.
- Run requirements discovery directly with client stakeholders, converting current-state operations into automation specifications, scope, and acceptance criteria.

Senior RPA Developer

FiiBo - Fortaleza, CE, Brazil | April 2024 - September 2025

- Delivered 20+ production RPA automations for healthcare provider operations, owning each end to end from process discovery and documentation through development, deployment, and support.
- Replaced manual, repetitive back-office workflows across multiple business fronts with reliable, scalable automation - removing over 500 hours of manual processing per month, automating tens of thousands of records monthly, and eliminating a class of manual data-entry error.
- Built AI agents and n8n automations running in production for internal teams and client accounts, integrating disparate systems to shorten delivery cycles.
- Developed automations in Python using Selenium, Pandas, and OpenCV to interact with web, desktop, and image-based interfaces where no API was available.
- Produced operational dashboards for automation monitoring and provided ongoing production support, giving stakeholders visibility into throughput and failures while hardening processes against upstream system changes.

RPA Analyst Developer

Cadmus IT Solutions - São Paulo, SP, Brazil | March 2022 - September 2022

- Automated employee onboarding and offboarding, provisioning and revoking system access across HR, IT asset management, and finance platforms.
- Delivered roughly six automations in seven months - three built new, three inherited unfinished or failing and rebuilt - working directly with process owners through development, testing, deployment, and support.
- Led the Python RPA squad's technical work, guiding one to two developers and up to three interns, and introduced the team's first standards for project structure, code review, and documentation.

Python Developer

BPA Technologies - São Paulo, SP, Brazil | December 2020 - March 2022

- Delivered end-to-end process automation into client operations across insurance, healthcare, financial services, logistics, and consumer products, at an RPA-as-a-service provider.
- Owned four to five automations from discovery through development, testing, deployment, and rollout on an in-house Python RPA framework, and handled production support: triaging tickets, resolving defects, and supporting key users.
- Automated web, desktop, and file-based workflows with Selenium, Pandas, Openpyxl, and OpenCV, integrated systems via REST and Google Workspace APIs, and built the SQL reporting and dashboards behind process status and failure visibility.

SELECTED PROJECTS

support-agent - Python, LangGraph, Anthropic API, Pytest, GitHub Actions

- Open-source support agent at github.com/feliperaro/support-agent, built to show production practice rather than a demo: a LangGraph classify-retrieve-act loop with hybrid retrieval, tool-level access control that refuses privileged actions before identity is verified, and output guardrails that block invented pricing.
- Paired it with an evaluation harness that scores behavior structurally - tool calls, refusals, trajectory order, retrieval grounding - rather than response wording, running on a fresh clone with no API key and gated in CI against a score floor.

EDUCATION

Postgraduate Degree, Artificial Intelligence for Developers

FIAP (Postech) - São Paulo, Brazil | August 2024 - June 2025

Associate Degree, Information Technology

Tamwood Careers College - Toronto, Canada | July 2023 - December 2023

Bachelor's Degree, Systems Analysis and Development

FIAP - São Paulo, SP, Brazil | February 2020 - December 2021

Technical Degree, Information Technology

Senac - Taboão da Serra, SP, Brazil | February 2019 - July 2020